Year 3 CSE - Lab 6

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Push Notifications with Firebase Cloud Messaging

Observations

- Firebase requires platform-specific config file for the Android (google-services.json).
- FCM behaves differently depending on app state (foreground/background/terminated).
- The FCM token is unique per device
- Foreground notifications don't auto-display on Android. need flutter_local_notifications

Key steps followed.

1. Created a Firebase project in the console.
2. Registered the Android using the application id from build.gradle.kts
3. Downloaded and placed google-services.json in android/app/
4. Added firebase_core and firebase_messaging to pubspec.yaml.
5. Configured build.gradle files (project & app level) with google-services plugin.
6. Initialized Firebase in main.dart with Firebase.initializeApp()
7. Requested notification permission and retrieved the FCM token.
8. Set up message handlers for ~~app~~ all 3 app states
9. Tested by sending a message from Firebase console using the FCM token.